



CCN Test Procedure

AD-300 - MHWIRTH (DLS)

Change No:

0006

CCN Title/Identification:

AD 300 hamonization update based on Commissioning findings up to AD 303

Application:

DWCS PLC Drawworks Control System - PLC

Test Description:

Software upgraded to TIA20
PLC has been upgraded to a higher firmware version.

Parameters that need to be checked and verified:

"PositionDb".slo.sHeight.sCalculation.sFlag.fFlagHeightMeasuredFromFunnelForLowering_RW

"PositionDb".slo.sHeight.sCalculation.sFlag.fFlagHeightMeasuredFromFunnel

"MainDriveDb".slo.sSpeedMonitor.fSpeedDeviationAllowed_RW

"PositionDb".slo.sLoopMonitoring.fDelayTimeOverspeedDetection_RW

Step by Step Procedure:

Step #	Description / What to do	Acceptance criteria / What to look for	Comments	Passed Site/By/Date
1	Update PLC firmware prior to installation of TIA20 version.	PLC firmware is updated.	Make sure to back-up the existing SW before starting.	RIG SIM
2	Upload SW with correct TIA version.	PLC is in run, IOs are connected, communication towards all connected nodes is established.		RIG SIM
3	Setup/verify height calculation.	Height calculation is operating correctly.	Setup according to DWCS support document. At least heights for Flag need to be corrected.	RIG SIM
4	Setup main drive speed monitoring.	Monitoring is setup, no alarm is initiated during normal operation.	Setup according to DWCS support document. Limit value should not exceed 200.	RIG SIM
5	Check function of all alarms from diagnostic for VFD and Braking Chopper.	Whenever one of the devices (VFD or BC) is faulty or not available a dedicated alarm should be indicated.		RIG SIM ACB 12.02.26
6	Indication of speed limitation with active reduced motor mode.	Limit indication in HH screens is updated when reduced motor mode is activated.		RIG SIM ACB 12.02.26
7	Check correct function of load limitation, when lowering with active limitation and directly changing joystick deflection to hoisting.	Lowering is possible and hoisting is blocked.		RIG SIM ACB 12.02.26

Administration / Sign Out:

Test Site	Accepted by Customer	Accepted
RIG	☐	☐
SIM	☐	☒ Yes Achim Braml (ACB) 12.02.2026



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8	Check function of dead man switch, release during movement and press again.	Speed shall reduce whenever the dead man switch is released.		RIG SIM ACB 12.02.26
9	Check speed measurement in conjunction with overspeed detection. Fault delay to be set to 0,5s.	Measured velocity should not contain high peaks, overspeed should not be triggered.	Trace PerformanceChartVerification can be used for monitoring.	RIG SIM ACB 12.02.26
10	Setup shutdown handling for lowering detection. Verify the system recognizes drops when the DW power is lost.	Brake is closed within a short time on power loss. All other shutdown szenarios are operating as before (with power at the motors and WPS).	ShutdownMainDrive shall be used for verification. This needs to be verified with wire attached, block should be secured by usage of hangoff.	RIG SIM ACB 12.02.26
11	Operate the DW in Normal with both and one motor.	No alarm from the motion observer and tracking monitoring should be activated.		RIG SIM ACB 12.02.26
12	Check indication of actual reevings in the HH-HMI.	Correct actual reevings are indicated.		RIG SIM ACB 12.02.26
13	Check correct function of max. ZMS lowering speed.	DW should operate within the set limit, in case a lower limit is active the lower limit should apply.	This may be simulated by setting an ADS ROP limit below the ZMS limit.	RIG SIM ACB 12.02.26

Test Observations, Comments, etc.:

Administration / Sign Out:

Test Site	Accepted by Customer	Accepted		
RIG	☐	☐		
SIM	☐	☒ Yes	Achim Braml (ACB)	12.02.2026